

Table 1

| Important log fields for Web analysis |
|---------------------------------------|
| Date and time                         |
| IP address                            |
| URL accessed                          |
| Protocol (HTTP or FTP)                |
| Browser type                          |
| OS type                               |
| Cookie                                |
| Site referrer                         |

Web analysers parse the details of those text files to carry out an analysis. For example, by sorting based on the source IP address, we can find out how many hits were generated by a particular Web client; or by intelligently sorting through the Web page file names, we can know which pages are hit the most. Based on the values in the browser field and OS type, it is easy to know the number of Windows machines running the IE browser or Firefox, Mac users running Safari, etc. As you can see, all this information is extremely useful to tune the website according to the users' experience, thus increasing traffic and leading to better business.

## Top 10 Web analyser tools

Given below is our list of top 10 tools for Web analysis for mid to large IT Web infrastructures. We selected these tools based on their popularity, deployment base, and simplicity to install, configure and put to use. The list contains a few tools that can perform on-the-fly Web analysis, which can help in troubleshooting website code-related problems.

**AWStats:** Though this is one of the first-generation tools, it is still widely used. Written in Perl, it works well on multiple platforms. A great feature of AWStats is that it supports virtually all popular Web servers' log formats, right from Microsoft IIS and Linux Apache to O'Reilly Web servers. It is capable of creating customisable views, including bar graphs and pie charts, thus offering a clear insight into Web traffic statistics. AWStats is meant for small to medium infrastructures, where log files are not too heavy to process. This tool is managed and updated at <http://awstats.sourceforge.net>.

**Webalizer:** Unlike various GUI-based tools, Webalizer is a complete command-line-operable utility, which makes it popular among Linux and UNIX administrators. It has its own small configuration language, which can be used to decide how the tool should read and parse the log files, and the fields in it. For example, configuring its *IgnoreSite* option with an internal IP address range can help get rid of internal Web traffic, and focus only on external hits. Due to its extensive command switches, it can be used in a scheduled job to perform the daily administrative tasks of looking into Web logs or automatically creating useful reports. This tool can be downloaded from <http://www.webalizer.org>.

**Piwik:** When many log files contain huge amounts of information, it becomes cumbersome and time-consuming to parse those. This needs a faster log parsing tool, and Piwik solves the problem. Besides just the typical Web analysis, Piwik comes with a set of plug-ins to enhance the reporting styles. For example, its GeoIP plug-in can be utilised to map source IP addresses in the logs to a particular country, state or city. While supporting multiple platforms, it has its own Python-based command interface to get the most in terms of reports. Today, many Web hosting facilities use Piwik, and also provide its customisable Web user interface to their customers as an offering. This tool can be found at <http://piwik.org>.

**OpenWebAnalytics:** Written in PHP and using MySQL as the back-end, this utility comes in handy especially when administrators want to collaboratively process the logs of multiple websites together. OpenWebAnalytics is capable of processing really large logs, and can optionally fetch those directly from a database format too. Unlike many other professional tools, this open source version can provide a click-stream report, whereby the users clicking on a Web page are shown in a date-and-time format. This helps code troubleshooters to know exactly what the Web user did, and they can try repeating those steps to replicate the problem. It can also create a heat-map type of report, segregating the website into most-hit and least-hit pages, shown in the form of colour gradients for easy understanding. This tool is available at <http://www.openwebanalytics.com>.

**Deep Log Analyser:** This tool is a typical Web analysis utility. However, unlike most other tools, it is very useful to process FTP logs. Besides the standard reporting, it can create a list of keywords and the hits on the Web pages that contain that keyword too. This is especially important for SEO (Search Engine Optimisation). The tool uses a standard MS-Access style database format, which can be exported to any other database engine and queried with standard SQL for further customisation of reports. For small infrastructures, Deep Log Analyser can be more than adequate to get visibility for a Web business. This tool can be found at <http://www.deep-software.com>.

**FireStats:** This tool comes equipped with features meant for a different audience. Today, most websites are being developed in open source content management systems such as Drupal, Joomla or WordPress. These systems have their own unique styles of handling file names, cookies and other parameters. FireStats is capable of interpreting cookies found in the Web log files, as well as the names of code files accessed by the user, and segregating information for each CMS in use. It can translate the report into multiple languages. Unlike many other tools, FireStats can be installed on the Web server, and it can act as a silent background